

Credit Card Fraud Detection

Exploratory Data Analysis (EDA) Summary

1. Project Objective

The objective of the EDA was to understand the structure of credit card transactions, identify data-quality issues, examine how fraudulent transactions differ from legitimate transactions, and identify patterns that could support fraud-monitoring and dashboard analysis. The EDA was intentionally kept business-focused and did not include predictive machine-learning modeling.

2. Dataset Overview

Metric	Result
Total transactions	284,807
Total columns	31
Legitimate transactions	284,315
Fraudulent transactions	492
Fraud rate	0.1727%
Missing values	0
Duplicate rows	1,081

3. EDA Process: What We Did and Why

A. Dataset inspection

Loaded the CSV and examined the first records, dimensions, column types, missing values, duplicates, and descriptive statistics. This established the dataset structure and confirmed that all 31 columns were numeric and that there were no missing values.

B. Target/class analysis

Compared legitimate and fraudulent transactions using Class. This was necessary because fraud is the central outcome of the project and the dataset is highly imbalanced.

C. Duplicate investigation

Investigated duplicated rows by fraud class before deciding what to do with them. There were 1,081 duplicated rows overall, with 1,822 legitimate and 32 fraudulent rows belonging to duplicated groups. Duplicates were retained rather than blindly deleted.

D. Transaction amount analysis

Compared transaction amounts between legitimate and fraudulent transactions using averages, medians, percentiles, amount bands, and distribution plots. This was done to understand whether transaction value was associated with fraud and whether fraud was concentrated in high-value transactions.

E. Amount-band analysis

Grouped transactions into €0-10, €10-50, €50-100, €100-500, €500-1,000 and €1,000+ bands. This allowed fraud rates to be compared within transaction-value ranges rather than relying only on the mean.

F. Time analysis

Converted the elapsed-seconds Time field into day and hour-of-day fields. Fraud counts and fraud rates were then examined by hour to identify temporal patterns.

G. PCA feature analysis

Compared the means of V1-V28 between legitimate and fraudulent transactions and calculated their correlations with Class. Because the variables are anonymized PCA components, they were treated as statistical signals rather than given business meanings.

H. Distribution visualization

Used boxplots for V17 and V14 to visually compare legitimate and fraudulent distributions. These plots confirmed substantial distribution shifts for these features.

I. Final business-focused visual analysis

Created transaction-amount distributions, fraud-rate-by-hour, and fraud-count-by-hour charts. These visuals connect the statistical findings to practical fraud-monitoring questions.

4. Key Findings and Results

Area	Key result
Class imbalance	492 frauds out of 284,807 transactions, or 0.1727%.
Amount	Fraud mean = €122.21; fraud median = €9.25. Legitimate mean = €88.29; median = €22.00
Small transactions	249 fraudulent transactions were in the €0-10 band, showing that fraud is not limited to expensive transactions
Highest amount-band fraud rate	€500-1,000 had the highest observed rate at 0.4192%.
Lowest amount-band fraud rate	€10-50 had the lowest observed rate at 0.0628%.
Time	02:00 had the highest observed hourly fraud rate at 1.7127%.
Fraud count by hour	02:00 had 57 fraud cases; 11:00 had 53.
PCA signals	V17, V14, V12, V10 and V16 had the strongest correlations with Class among the PCA variables examined
V17 distribution	Fraudulent transactions showed a strong downward shift relative to legitimate transactions
V14 distribution	Fraudulent transactions also showed a strong downward shift relative to legitimate transactions

5. Which Factors Show the Strongest Association with Fraud?

Within the available anonymized data, the strongest linear associations with the fraud label were observed for V17 (0.326), V14 (0.303), V12 (0.261), V10 (0.217), V16 (0.197), V3 (0.193), V7 (0.187), V11 (0.155), V4 (0.133), and V18 (0.111).

These values describe association, not causation. The V1-V28 fields are PCA-transformed and anonymized, so they cannot be interpreted as specific business attributes such as income, merchant type, or location.

6. Final EDA Outcomes

- Fraud is extremely rare, creating a severe class-imbalance problem.
- Fraud occurs across transaction values and is not simply a high-value transaction problem.
- The fraud amount distribution is strongly skewed: many frauds are small, while the upper tail is higher than for legitimate transactions.
- Fraud rates vary substantially by hour in the two-day observation period.
- Several PCA components show strong differences between legitimate and fraudulent transactions, especially V17 and V14.
- Transaction amount and time provide useful business-level segmentation, while the anonymized PCA components provide strong statistical signals.
- The findings support building a Power BI dashboard focused on fraud volume, fraud rate, transaction amount patterns, time patterns, and PCA-feature comparisons.